TECHNICAL ASSIGNMENT

SamaSocial

Full-Stack AI Feature Build

Role	AI/Full-Stack Developer
Deadline	3 Days from assignment receipt
Submission	GitHub repo link + short Loom/video demo
Stack (preferred)	FastAPI · React / Next.js · Supabase · any LLM API

Overview

Samasocial (samasocial.in) is a social learning platform connecting learners and educators. We are actively expanding our AI capabilities and this assignment is designed to evaluate your ability to build production-quality AI features.

This assignment has two independent tasks. You may attempt one or both — completing both tasks will be weighted more heavily during evaluation.

Task 1 — Multi-Source AI Learning Assistant

What to Build

Build a web-based AI chatbot that accepts one or more knowledge sources from the user, processes them, and then answers questions, explains concepts, and resolves doubts based on that content.

Supported Input Sources

- YouTube video URL — transcribe or summarise the video and use it as context
- PDF file — extract and chunk text, build a retrieval index
- PowerPoint / PPTX file — parse slides, extract text + structure
- Any public webpage URL — scrape and parse the page content

The user should be able to mix and combine sources in a single session (e.g. a PDF + a YouTube video together).

Chatbot Behaviour

- Answer questions grounded strictly in the uploaded/linked content
- Explain topics in simple language on request ("explain this in simple terms")
- Handle cross-questions and follow-up queries in the same session
- Cite or reference the source when answering (e.g. "from slide 4" or "at 3:22 in the video")
- Gracefully decline questions that are out of scope of the provided material

Technical Requirements

1. Chunking & Retrieval — use vector embeddings or keyword search to retrieve relevant chunks before calling the LLM. Do not dump the full document into a single prompt.
2. Streaming responses — the chatbot should stream the reply token by token, not wait for the full response.
3. Session memory — the conversation history should persist for the duration of the session so follow-up questions work naturally.
4. Basic UI — a clean chat interface; file upload area; source badges showing what has been loaded.

Bonus Points

- Support multiple simultaneous sources and indicate which source each answer came from
- Add a "quiz me" mode that auto-generates questions based on the loaded content
- Show a short summary of each source once it has been processed

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Task 2 — AI Course Planning Assistant for Mentors

What to Build

Build a conversational AI assistant designed specifically for mentors and educators. The assistant helps a mentor plan a complete, well-structured course for their students through a guided back-and-forth conversation.

Core Functionality

- Intake — the assistant asks the mentor key questions to understand the course:
  - Subject / topic area
  - Target audience (age group, skill level, prior knowledge)
  - Duration and session frequency
  - Learning goals and outcomes
- Course generation — based on the mentor's answers, generate a structured course plan:
  - Module breakdown with titles and learning objectives
  - Suggested lesson topics per module

    - Recommended resources per lesson — publicly available only (YouTube videos, blog posts, articles, documentation, and practice exercises from platforms like HackerRank, LeetCode, Kaggle, etc.) Suggested assessments or quizzes at module end
- Refinement — the mentor can ask follow-up questions to adjust any part of the plan (e.g. "make module 2 simpler" or "add a project-based assignment")
- Export — the final course plan should be:
    - Exportable as structured JSON for system integration
    - Viewable as a live preview directly in the web UI, updating in real time as the mentor refines the plan

Technical Requirements

5. Multi-turn conversation — maintain the full context of the planning session across turns.
6. Structured output — the final course plan must be produced as structured data (JSON), not just free-form text, so it can be consumed by an external system.
7. Editable output — once the plan is generated, the mentor should be able to click and edit individual fields directly in the UI.
8. Clean UI — a split-panel design works well: chat on one side, live course plan preview on the other.

Bonus Points

- Allow the mentor to paste an existing syllabus or curriculum PDF and have the assistant improve or restructure it
- Add a difficulty progression indicator per lesson (beginner / intermediate / advanced)
- Suggest prerequisite topics the student should know before each module

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Evaluation Criteria

Both tasks will be judged on the following dimensions:

Criterion	What We Look For	Weight
AI Quality	Accurate, relevant, grounded responses; good retrieval; no hallucination	**30%**
Code Quality	Clean, readable, well-structured code; sensible abstractions	**25%**
UI / UX	Intuitive interface; responsive; thoughtful error states and loading UX	**20%**
Architecture	Separation of concerns; scalable design; environment config handled properly	**15%**
Bonus Features	Any bonus items attempted and working	**10%**

Submission Guidelines

What to Submit

- A public GitHub repository with all source code
- A README.md with: setup instructions, environment variables needed, and architectural decisions made
- A short demo video (3–5 minutes) — Loom or any screen recorder is fine — showing tasks working end-to-end
- A deployed live link is a strong bonus but not required

Rules

- You may use any LLM API (OpenAI, Anthropic, Groq, Gemini, etc.) and any open-source libraries
- Do not use fully pre-built chatbot SaaS products (e.g. Botpress, Voiceflow) as the core — the logic must be yours
- Commit history matters — we want to see how you build, not just the final result
- If you get stuck on something, document it in the README rather than silently skipping it

Questions & Contact

If you have any questions about the requirements or need clarification, reach out before starting — not on the last day. We are happy to help you understand the scope.

We are not looking for perfection. We are looking for clarity of thinking, good engineering instincts, and the drive to ship.

Good luck!